LipReadingAPP

From Visual Speech to text

GROUP 5

Lip Reading App Team

Guidance: Prof Max

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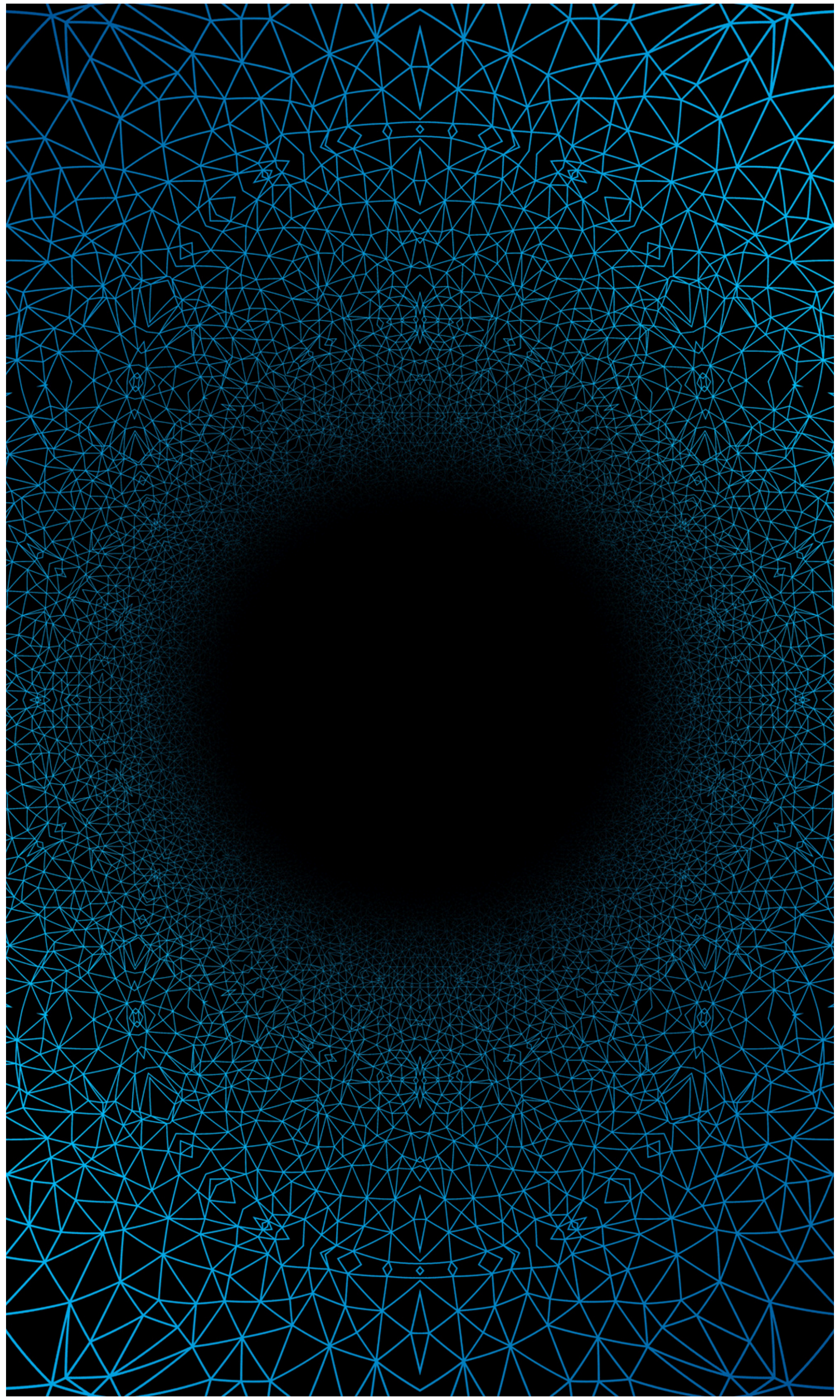
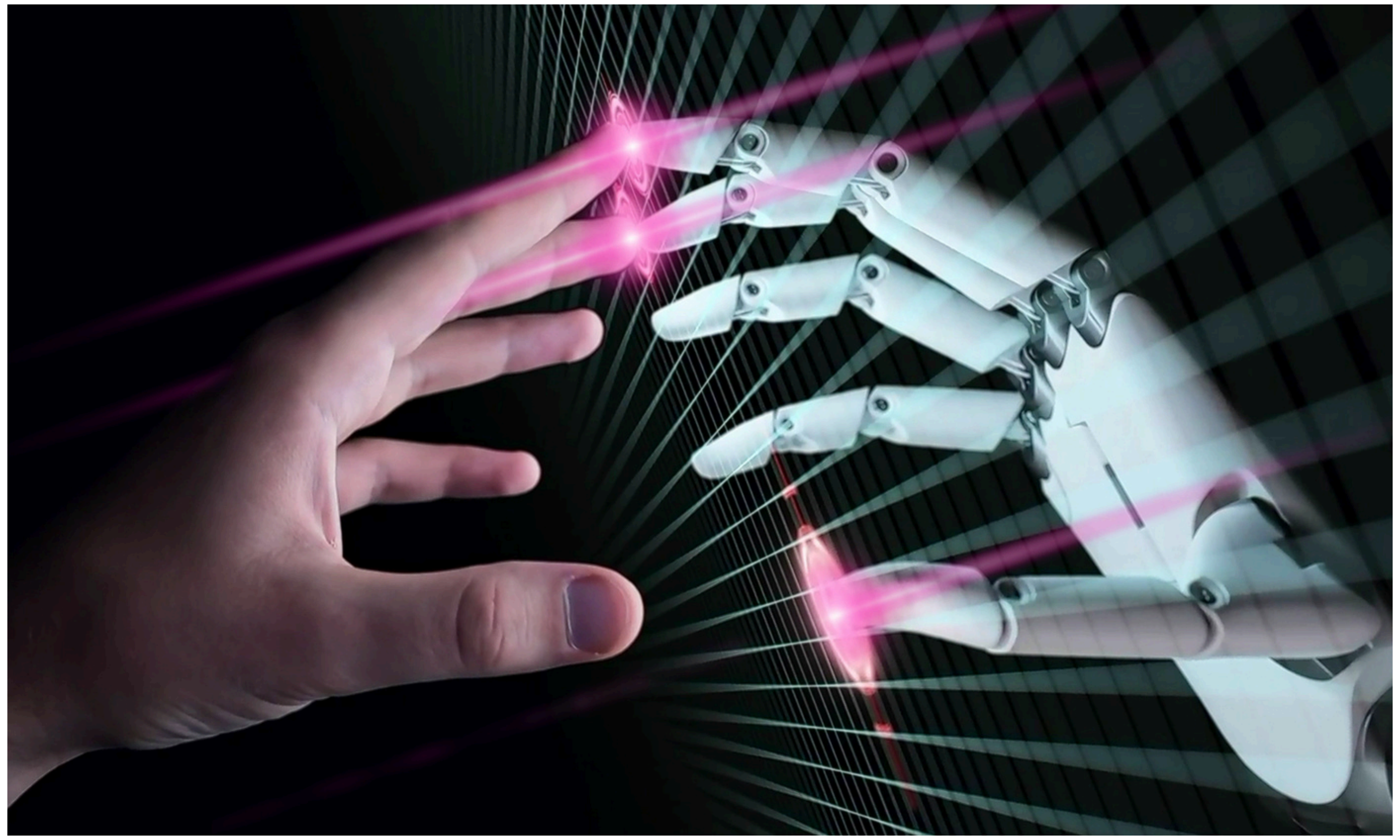


INTRODUCTION

- Speech is fundamental to human interaction, yet there are many environments where audio cannot be relied upon — due to noise, technical limitations, or disabilities.
- **Visual speech recognition** (commonly referred to as lip reading) is the process of understanding spoken words purely from the visual cues of a speaker’s lips, without any auditory input.
- In this project, we developed a **deep learning-based lip reading system** capable of **decoding silent videos into meaningful text**, thereby bridging the gap where traditional audio-based communication fails.
- Our solution leverages spatio-temporal modeling techniques, combining the power of **3D Convolutional Neural Networks (3D CNNs)** for feature extraction and **Bidirectional LSTM networks** for sequence modeling, enhanced with **CTC loss** for dynamic alignment between frames and characters.

Problem Statement



Despite recent advances in speech recognition, most systems depend heavily on audio signals. However, in real-world scenarios such as noisy environments, surveillance without sound, or communication for the hearing-impaired, reliable audio is often absent or unusable.

The core problem we address is:
"Can we accurately predict the sequence of spoken words from only a silent video, purely by analyzing lip movements?"

01 To solve this, the model must:

- Precisely detect and focus on the mouth region across video frames.
- Extract complex spatial and temporal features from sequences of lip movements.
- Understand the sequence of phonetic articulations over time without explicit manual labeling.
- Align varying video lengths and speaking speeds automatically to corresponding textual outputs.

02 Challenges faced include:

- Variations in lip shapes, speaking styles, and frame rates.
- Blurry or low-quality video frames.
- Unaligned frame-to-character mappings, making supervised learning more complex.
- Achieving end-to-end training without needing detailed frame-by-frame word alignments.

Why This Problem Matters

01 **Accessibility**
Helping the hearing-impaired engage in more natural communication.

02 **Silent Speech Interfaces**
Enabling interaction where speaking aloud isn't possible (e.g., military, underwater, space missions).

03 **Security and Surveillance**
Analyzing silent video footage to reconstruct conversations.

04 **Noise-Robust Communication Systems**
Providing alternative communication modes in extremely noisy environments like factories or battlefields.

Dataset Used

For training and validating our lip-reading model, we used the GRID Corpus Dataset, a widely recognized benchmark for visual speech recognition research.

Dataset Name: GRID Corpus
<command> <color> <preposition> <letter> <digit> <adverb>
eg. bin blue at f two now
Purpose: Training and evaluation of lip-reading models, notably used for developing LipNet.

Data Volume:

1. 1,000 processed video samples from each speaker.
2. Each video consists of 75 frames (around 3 seconds duration).
3. Drawn from 34 different speakers, providing diversity in speaking styles and lip movement patterns.

Technology Stack & What They Do

01 MTCNN (Multi-task Cascaded Convolutional Networks)

- Detects faces and localizes key landmarks (eyes, nose, mouth).
- Used to extract the mouth region from raw video frames.
- Works well even on low-resolution or slightly blurry frames.

02 3D CNN (Spatio-Temporal Feature Extraction)

- Applies 3D filters across width, height, and time.
- Captures both appearance and motion across frames.
- Essential for learning lip movement patterns.

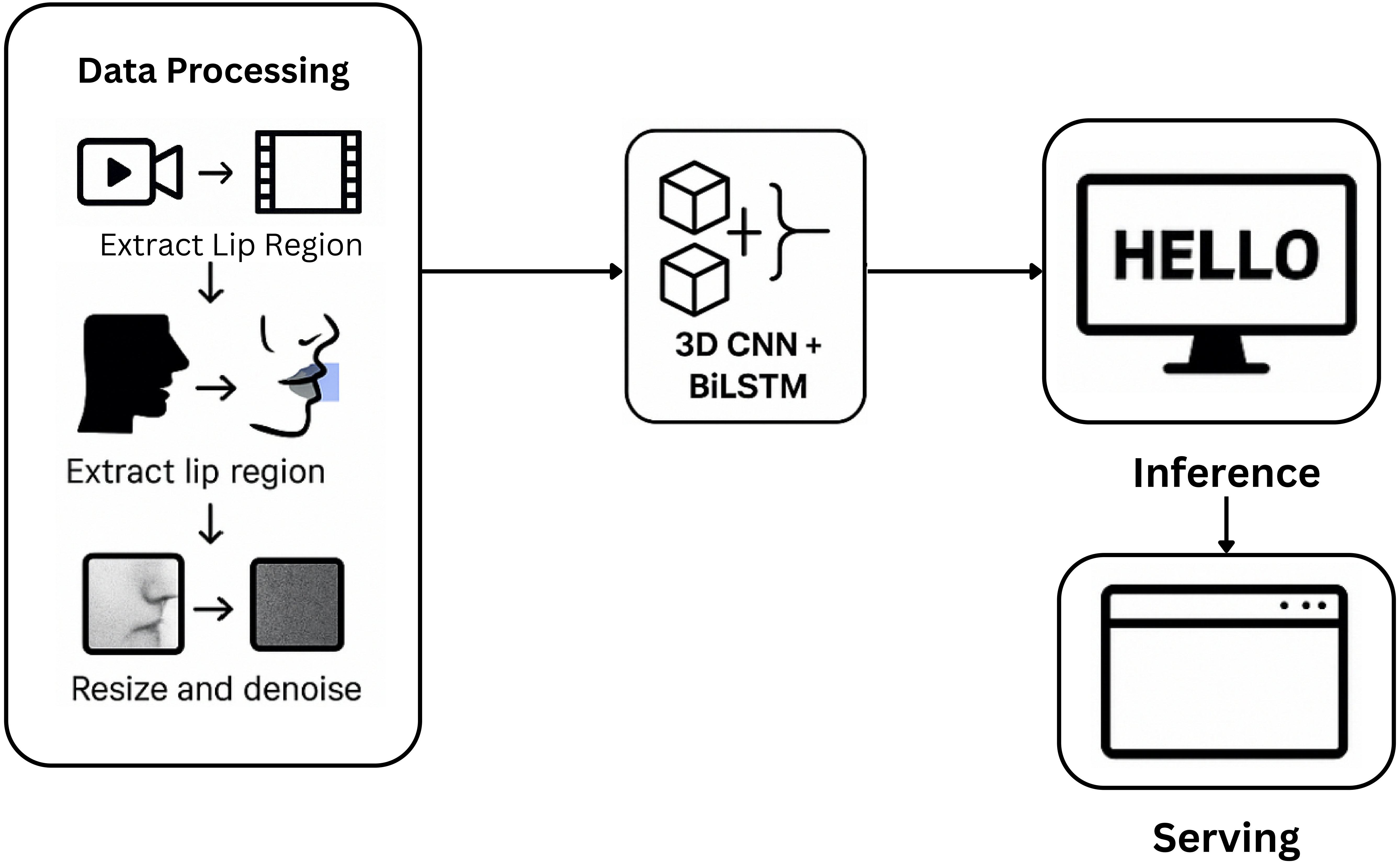
03 Bidirectional LSTM (Long Short-Term Memory Networks)

- Processes features from 3D CNN in both forward and backward directions.
- Handles temporal dependencies and speaking speed variations.
- Improves word prediction by using context from both directions.

04 CTC Loss (Connectionist Temporal Classification)

- Enables training without exact alignment between frames and text.
- Learns how to align visual input with character output.
- Solves the variable length challenge in lip reading.

Workflow Design



Model Architecture

Layer (type)	Output Shape	Param #
conv3d (Conv3D)	(None, 75, 50, 150, 128)	10,496
batch_normalization (BatchNormalization)	(None, 75, 50, 150, 128)	512
activation (Activation)	(None, 75, 50, 150, 128)	0
max_pooling3d (MaxPooling3D)	(None, 75, 25, 75, 128)	0
conv3d_1 (Conv3D)	(None, 75, 25, 75, 256)	884,992
batch_normalization_1 (BatchNormalization)	(None, 75, 25, 75, 256)	1,024
activation_1 (Activation)	(None, 75, 25, 75, 256)	0
max_pooling3d_1 (MaxPooling3D)	(None, 75, 12, 37, 256)	0
conv3d_2 (Conv3D)	(None, 75, 12, 37, 256)	1,769,728
batch_normalization_2 (BatchNormalization)	(None, 75, 12, 37, 256)	1,024
activation_2 (Activation)	(None, 75, 12, 37, 256)	0
max_pooling3d_2 (MaxPooling3D)	(None, 75, 6, 18, 256)	0
spatial_dropout3d (SpatialDropout3D)	(None, 75, 6, 18, 256)	0
time_distributed (TimeDistributed)	(None, 75, 27648)	0
bidirectional (Bidirectional)	(None, 75, 1024)	115,347,456
dropout (Dropout)	(None, 75, 1024)	0
bidirectional_1 (Bidirectional)	(None, 75, 1024)	6,295,552
dropout_1 (Dropout)	(None, 75, 1024)	0
dense (Dense)	(None, 75, 29)	29,725

Total params: 124,340,509 (474.32 MB)
Trainable params: 124,339,229 (474.32 MB)
Non-trainable params: 1,280 (5.00 KB)

Model Results

Dataset Split	Speakers	Purpose	WER (%)
Train	6 speakers (80%)	Model learns features	—
Validation (Seen)	10% from same 6	Tune hyperparameters	19.6
Test (Seen)	Remaining from same 6	Evaluate generalization on seen data	18.9
Test (Unseen)	2 new speakers	Evaluate speaker-independence	34.7

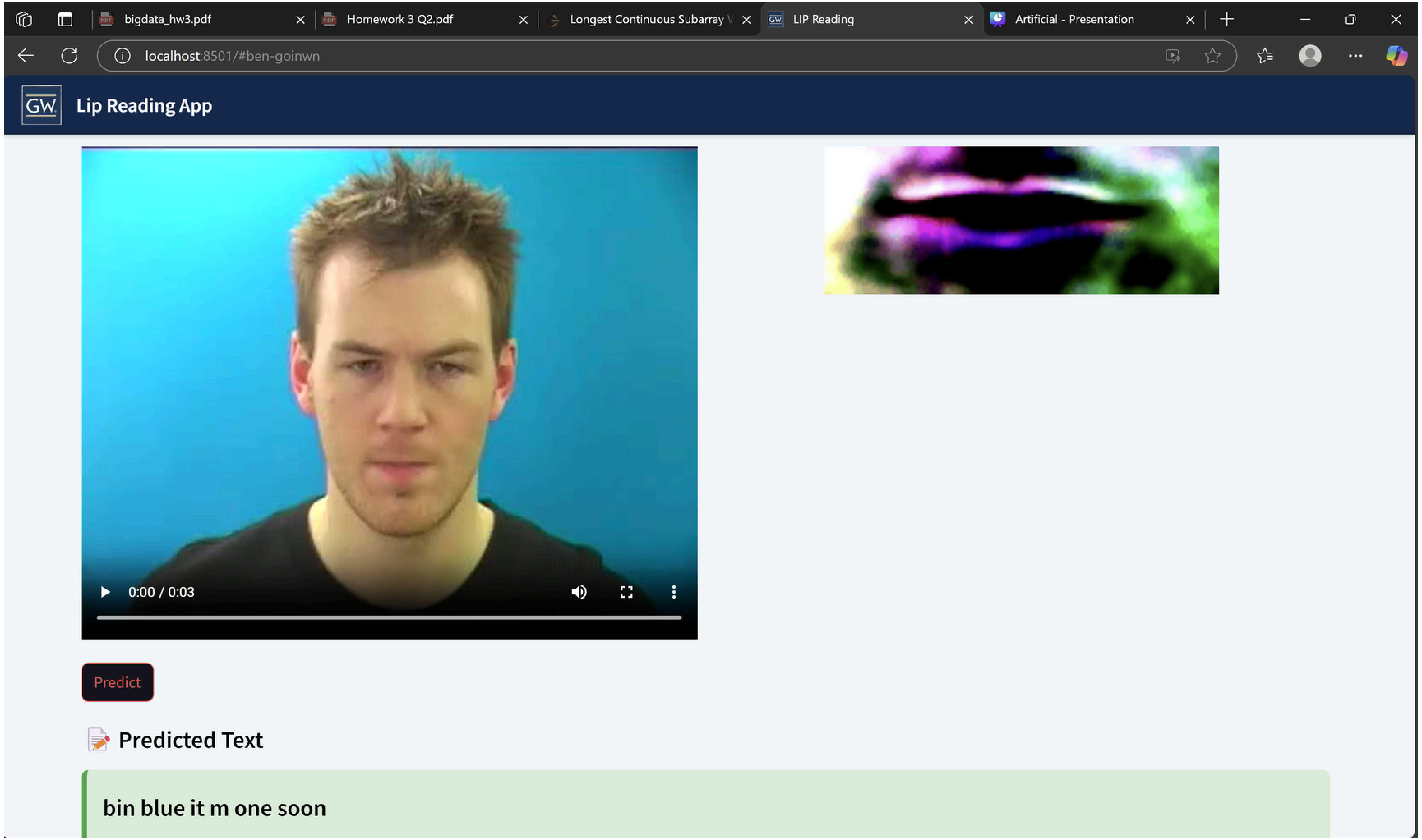
Model Results Over Epochs

```
Epoch 1/100
1/1 ━━━━━━━━━━━━━━━━━━━━ 3s 3s/step
Original: place red with a eight please
Prediction: l e e e o
202/202 ━━━━━━━━━━━━━━━━━━━━ 435s 2s/step - loss: 98.6251 - val_loss: 70.4242 - learning_rate: 1.0000
Epoch 2/100
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 63ms/step
Original: set white by f seven soon
Prediction: la e e t o o
202/202 ━━━━━━━━━━━━━━━━━━━━ 121s 597ms/step - loss: 70.6299 - val_loss: 63.2937 - learning_rate: 1.0
Epoch 3/100
```

```
202/202 ━━━━━━━━━━━━━━━━━━━━ 120s 595ms/step - loss: 9.4646 - val_loss: 14.4735 - learning_rate: 1.000
Epoch 29/100
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 61ms/step
Original: place red with n four now
Prediction: place red with four now
202/202 ━━━━━━━━━━━━━━━━━━━━ 120s 596ms/step - loss: 8.8243 - val_loss: 12.9147 - learning_rate: 1.000
Epoch 30/100
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 63ms/step
Original: place blue by f nine soon
Prediction: place blue by five son
```

```
Epoch 87/100
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 64ms/step
Original: place blue by m four please
Prediction: place blue by m four please
202/202 ━━━━━━━━━━━━━━━━━━━━ 121s 597ms/step - loss: 2.3469 - val_loss: 17.1131 - learning_rate: 3.346
Epoch 88/100
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 62ms/step
Original: lay red in u zero now
Prediction: lay red at u zero now
```

Final Output



Conclusion

- In this project, we successfully designed and implemented an end-to-end deep learning system capable of predicting spoken words from silent videos.

- By combining powerful models — MTCNN for precise mouth detection, 3D CNNs for spatio-temporal feature extraction, Bidirectional LSTMs for sequential modeling, and CTC loss for flexible alignment — we built a robust pipeline for visual speech recognition.

- Additionally, we made our solution accessible and interactive through a Streamlit-based web application, allowing real-time testing and showcasing the model's capabilities to users.

- Future extensions could explore:

1. Training on more complex like LRS2 and LRS3
2. Integrating attention mechanisms to improve focus on key frames.
3. Expanding to continuous sentence-level predictions for more natural conversation modeling.
4. Overall, this project is a step toward building intelligent systems that can hear through sight — opening new avenues where sound is not enough.
5. Making an interview tracking system for lip sync detection capturing specific video length and matching the real transcript.

Thank You